EXP. NO.: 02
DATE: 03.02.26
PG. NO.: 03

DETERMINATION OF CURRENT IN CIRCUIT USING MESH ANALYSIS

AIM:

To determine the current in circuit using mesh analysis both theoretically and practically for a given DC circuit.

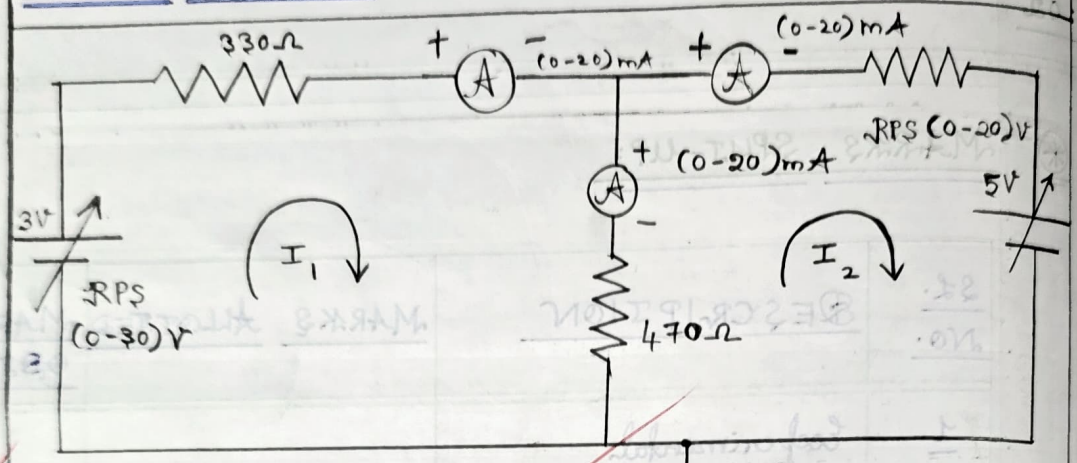
APPARATUS REQUIRED:

SL. NO.	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power Supply (RPS)	(0-30 V)	1
2.	Ammeter	(0-10 mA) MC	3
3.	Resistors	330Ω, 470Ω, 560Ω	Each 1
4.	Bread board	–	1
5.	–	–	–

PROCEDURE:

1. Give connections as per the circuit diagram.
2. Switch ON the supply, vary the RPS (Regulated Power Supply) and set a particular input voltage.
3. Note down the readings of ammeters and voltmeters and tabulate them.
4. Vary the RPS for different input voltages and note down the readings of all the meters.
5. Reduce the RPS to its minimum value and switch OFF the supply.
6. Using the tabulated values, verify Kirchhoff's laws practically, and verify it theoretically.

CIRCUIT DIAGRAM



MESH - CIRCUIT

⊛. TABULAR COLUMN:

PARAMETERS	THEORETICAL	PRACTICAL
I_1	1.2 mA.	0.5 mA.
I_2	4.2 mA.	4.8 mA.
$(I_1 - I_2)$	5.4 mA.	5.4 mA.

⊛. CALCULATIONS:

THEORETICAL CALCULATION:

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$R_1 = 330\Omega + 470\Omega = \underline{800\Omega}$.

$R_{12} = 470\Omega$; $R_{21} = \underline{470\Omega}$.

$R_{22} = (560 + 470)\Omega = \underline{1030\Omega}$.

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$\Delta = (800 \times 1030) - (470)^2$
$= \underline{603100}$.

$$\Rightarrow \Delta_1 = \begin{vmatrix} 3 & -470 \\ -5 & 1030 \end{vmatrix} = 3090 - 2350 = \underline{740}.$$

$$\Rightarrow \Delta_2 = \begin{vmatrix} 800 & 3 \\ -470 & -5 \end{vmatrix} = -4000 + 1410 = \underline{-2590}.$$

- $I_1 = \dfrac{\Delta_1}{\Delta} = \dfrac{740}{603100} = \underline{1.2\,mA}$.
- $I_2 = \dfrac{\Delta_2}{\Delta} = \dfrac{-2590}{603100} = \underline{-4.2\,mA}$.
- $(I_1 - I_2) = 1.2 - (-4.2) = \underline{5.4\,mA}$.

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SL.No.	DESCRIPTION	MARKS ALLOTTED	MARKS OBTAINED
1.	Experimental Setup/ Connections	20 MARKS	20
2.	Observation/Readings	20 MARKS	20
3.	Calculation/Results	20 MARKS	20
4.	Viva	20 MARKS	20
5.	Record	20 MARKS	20
		TOTAL :	100

RESULT:

Thus, the Mesh analysis is verified practically and theoretically. The resultant currents for 5 / 3 V supply are:

(a). The current across 330 Ω resistor is : 1.2 A.

(b). The current across 560 Ω resistor is : 4.2 A.

(c). The current across 470 Ω resistor is : 5.4 A.

100
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11/2/26